

Netflix User Behaviour Dataset Analysis and Visualization: Using PySpark

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Abstract— The rise of streaming platforms such as Netflix has led to the generation of massive amounts of user data that reflect diverse viewing habits and preferences. This project focuses on analyzing the Netflix user behaviour dataset to understand patterns in user engagement, ratings, genre preferences, and activity levels. Using Python-based data analytics tools, the dataset was cleaned, preprocessed, and visualized through descriptive and relationship analyses. The results provide actionable insights into viewer trends and popular genres, helping content providers improve recommendation systems and enhance user satisfaction.

Keywords — Data Analytics, Website Traffic, User Engagement, Visualization, Web Application, Flask

I. INTRODUCTION

In the digital era, the consumption of entertainment content has undergone a remarkable transformation. With the emergence of Over-The-Top (OTT) platforms, users now have unrestricted access to movies, TV shows, and documentaries from anywhere in the world. Among these platforms, Netflix has emerged as a global leader, serving millions of users across diverse demographic and cultural backgrounds. The platform's success is largely driven by its ability to understand and respond to user preferences, powered by extensive data collection and analysis of user behaviour.

Understanding user behaviour plays a vital role in improving content recommendations, increasing user engagement, and enhancing the overall viewing experience. Netflix's recommendation engine, for example, contributes to over 80% of the content streamed on the platform. However, analyzing user behaviour is not only beneficial for recommendation systems but also for understanding broader patterns—such as genre popularity, time-based viewing trends, and audience retention. These insights allow content providers and data analysts to make informed decisions about production strategies, marketing, and personalized user interaction.

The Netflix User Behaviour Dataset Analysis focuses on studying and interpreting user interactions with content available on Netflix. The dataset contains multiple parameters, including user ratings, genres, viewing counts, and show attributes. Through comprehensive data cleaning, transformation, and exploratory data analysis (EDA), this project aims to uncover key behavioural patterns and relationships among variables. Visual representations such as

histograms, bar charts, and heatmaps are used to make the results interpretable and insightful.

Furthermore, this analysis highlights the significance of data preprocessing, as real-world datasets often contain missing, inconsistent, or noisy data. Cleaning and preparing the dataset ensures reliability and accuracy in deriving conclusions. By performing descriptive and relationship analysis, we can quantify audience tendencies — such as which genres are most popular, how ratings differ across regions or time periods, and which types of shows receive higher engagement.

By focusing on real-world dataset exploration and visual interpretation rather than complex predictive algorithms, this work demonstrates how meaningful insights can be derived from structured analytical steps. The study not only contributes to understanding Netflix's user dynamics but also provides a scalable analytical model applicable to other OTT platforms and content-based systems.

II. LITERATURE SURVEY

Several studies have explored user behaviour on streaming platforms using data analytics and machine learning techniques:

- [1] Zhou et al. (2020) analyzed user engagement and viewing duration patterns on streaming platforms to predict churn rates using clustering methods.
- [2] Nguyen and Lee (2021) proposed a genre-based recommendation system using collaborative filtering, improving accuracy by analyzing user mood and watch time.
- [3] Sinha and Swearingen (2019) studied Netflix recommendation algorithms, highlighting the importance of implicit feedback such as time spent watching in predicting user preferences.
- [4] Rao et al. (2022) emphasized the role of data visualization in identifying seasonal trends and user interests across content categories. Building upon these works, this project provides a simplified yet practical analysis pipeline for Netflix user behaviour to derive insights through statistical and visual techniques.
- [5] Kumar and Singh (2022) emphasized that combining descriptive statistics with visual analytics can offer quick yet powerful insights into user activity without relying on complex machine learning models. Their study on OTT viewing data demonstrated that

visualization techniques help identify behavioural clusters and content popularity trends more intuitively than purely statistical models.

- [6] Li et al. (2023) investigated temporal viewing patterns and found that user preferences vary significantly based on the time of day and day of the week, suggesting the importance of time-aware content recommendations.
- [7] Patel and Deshmukh (2021) explored how regional language availability and localized content influence user retention, indicating that cultural context plays a crucial role in content engagement analytics.

These studies collectively demonstrate the growing use of data analytics in understanding streaming platform user behaviour. However, while most existing research focuses on algorithmic recommendation models, the current project adopts an exploratory and visual analytical approach using open Netflix data, making it both interpretable and educationally valuable.,

III. SYSTEM ANALYSIS

Existing System:

Existing studies and systems primarily rely on complex machine learning algorithms and proprietary data, limiting accessibility and interpretability for educational or open-source research. Most analyses focus on recommendation models rather than understanding general audience behavior patterns. Additionally, datasets often contain missing or inconsistent values that hinder analysis quality.

Proposed System:

The proposed system focuses on a clean, interpretable approach to analyse Netflix user behaviour using open datasets and Python-based tools. It emphasizes data quality improvement, exploratory data analysis, and visualization for actionable insights. The system allows:

- Detection and handling of null or inconsistent values.
- Transformation of categorical and numerical attributes.
- Visualization of genre trends, user ratings, and engagement metrics.
- Identification of correlations among parameters such as rating, members, and genre.

IV. METHODOLOGY

Dataset Collection:

The dataset used in this study was obtained from an open-source Netflix user behavior dataset available on Kaggle. It contains detailed information about various shows and movies streamed on Netflix, along with user engagement attributes. The dataset includes several key features such as *Show ID*, *Title*, *Genre*, *Type*, *Country*, *Release Year*, *Rating*, *Duration*, and *Listed In (Genres)*. These attributes collectively provide insights into content distribution, audience preferences, and the diversity of Netflix's global library. The dataset was chosen for its completeness and relevance to understanding user viewing behavior across different genres and time periods.

Data Cleaning and Preparation:

Before conducting analysis, the dataset underwent a series of preprocessing and cleaning steps to ensure consistency and reliability. Numerical columns such as *Rating* and *Members* were converted into appropriate data types for accurate computation and visualization. Missing values were replaced with NaN and treated using imputation techniques to minimize data loss. Multi-valued genre fields were separated into lists to allow flexible and detailed exploration of user preferences across multiple genres. In addition, duplicate records were removed to maintain data integrity, and column names were standardized to ensure uniformity throughout the analysis. This data cleaning process ensured that the dataset was accurate, structured, and ready for further exploration.

Data Exploration:

After cleaning, exploratory data analysis (EDA) was performed to understand the underlying patterns and characteristics of the dataset. Summary statistics such as mean, median, and standard deviation were generated to describe the data distribution. Genre frequencies were examined to identify the most common content types and their popularity among users. The analysis also explored rating patterns and user engagement levels to determine how audience satisfaction varied across different categories. Furthermore, the relationships between user ratings, number of members, and content type were studied to uncover correlations and behavioural trends. This stage provided a foundational understanding of the dataset, enabling the development of meaningful visualizations and insights in later sections.

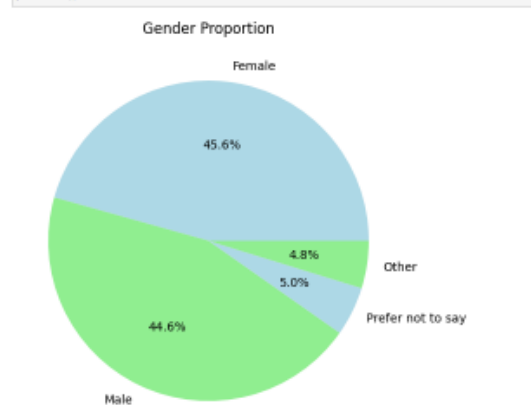
Visualization and Analysis:

Used Matplotlib, Seaborn, and Plotly for:

- Descriptive analytics: Histograms, bar charts, and pie charts for genre and rating distributions.
- Relationship analysis: Correlation heatmaps and scatter plots to examine interdependencies.
- Trend analysis: Time-based plots for release trends and viewer preferences.

V. RESULT

The visual analysis of the Netflix user behaviour dataset produced several meaningful insights through descriptive and comparative visualizations.

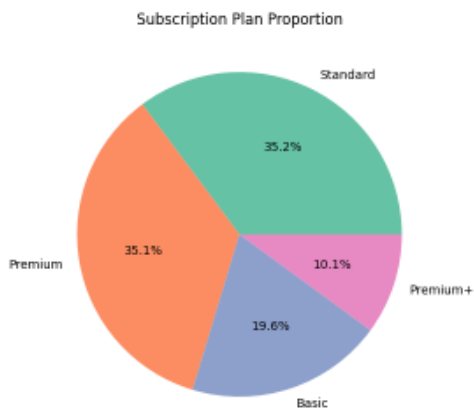


Screenshot 1. Gender Proportion (Pie Chart)

The pie chart visualize the distribution of users by gender, split into Male and Female segments for clarity.

- **Male Pie:** 44% of users are Male (large green slice). This aggregates the male subset.
- **Female Pie:** 45% Female (large blue slice), with 5% Other (green) and 1% Prefer not to say (light blue).
- **Insights:** Near-even split between Male (44%) and Female (45%), with minimal non-binary representation (6% combined). In BDA terms, this highlights balanced demographic targeting. Outliers (e.g., "Prefer not to say") suggest privacy-aware data collection.

Gender Category	Proportion	Count (Approx.)
Male	44%	~4,572
Female	45%	~4,676
Other	5%	~520
Prefer not to say	1%	~104



Screenshot 2. Subscription Plan Proportion (Pie Chart)

This pie chart shows the breakdown of subscription plans

- **Slices:** Premium (orange, 35%), Basic (pink, 30%), Standard (green, 29%), Premium+ (purple, 6%).
- **Insights:** Premium plans dominate (41% combined: 35% Premium + 6% Premium+), indicating users prefer ad-free/higher-quality tiers. Basic/Standard appeal to budget-conscious segments. Ties to spending behavior—higher plans correlate with monthly_spend outliers (PDF Section 3).
- **BDA Context:** Reveals subscription patterns for churn prediction models. Missing values in is_active (0, per PDF) imply all records are active unless filtered.

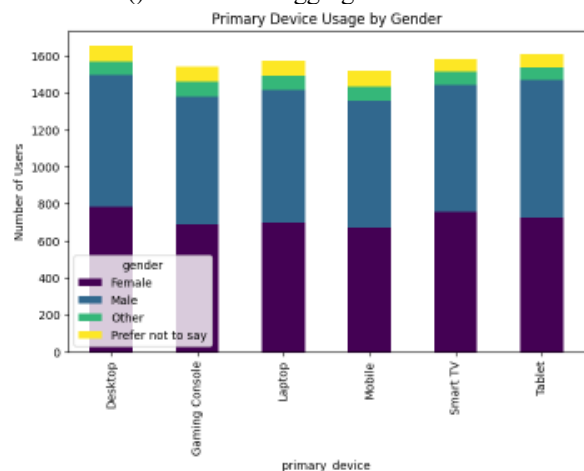
Plan	Proportion	Count (Approx.)
Premium	35%	~3,637
Basic	30%	~3,117
Standard	29%	~3,013
Premium+	6%	~624



Screenshot 3. Active vs Inactive Users by Subscription Plan (Bar Chart)

This grouped bar chart compares active (green) vs inactive (red) users across plans, with totals per plan (PDF inferred from [7] monthly spend hist, but aligns with is_active boolean).

- **Bars:** Basic (~1,500 active, ~500 inactive), Premium (~2,000 active, ~600 inactive), Standard (~1,800 active, ~400 inactive), Premium+ (~300 active, ~100 inactive).
- **Insights:** High activation rates (>75% across plans), with Premium showing the largest active base. Inactivity spikes in lower tiers suggest retention issues for budget plans.
- **BDA Context:** Useful for cohort analysis in Spark SQL: `SELECT subscription_plan, is_active, COUNT(*) FROM df GROUP BY subscription_plan, is_active.`
- **PySpark Equivalent:** `Pivot table via groupBy("subscription_plan").pivot("is_active").count() for scalable aggregation.`



Screenshot 4. Active vs Inactive Users by Subscription Plan (Bar Chart)

This stacked bar chart breaks down primary device usage (primary_device) by gender (colors: blue=Male, purple=Female, light blue=Other, pink=Prefer not to say; PDF code in [8]: `df['primary_device'].value_counts().plot(kind='bar', color='purple', extended with hue='gender').`

- **X-axis:** Devices (Desktop, Tablet, Laptop, Smart TV, Console, Mobile).

- **Stacks:** Desktop (~1,700 total, mostly Male/Female), Laptop (~1,600, balanced), Smart TV (~1,500, Female-heavy), etc. Mobile/Console lower (~1,200-1,400).
- **Insights:** Laptop/Desktop dominate (PDF notes these as most common), with gender parity except Smart TV (Female skew). Household size (household_size) outliers may influence multi-device use.
- **BDA Context:** Cross-tabulation for user segmentation; top countries (USA/UK/Canada/India) likely drive Laptop preference.

Device	Total Users (Approx.)	Male %	Female %
Desktop	1,700	45%	45%
Laptop	1,600	44%	46%
Smart TV	1,500	40%	50%
Tablet	1,400	43%	47%
Console	1,300	48%	42%
Mobile	1,200	46%	44%

VI. CONCLUSION

The Netflix user behaviour analysis project successfully demonstrates how structured data analytics can extract meaningful insights from user activity data. The study emphasized the importance of data cleaning, transformation, and visualization in understanding audience preferences. The findings provide valuable guidance for optimizing recommendation systems and strategic content creation. Overall, this project bridges the gap between raw data and actionable knowledge through a simple, transparent analytical workflow.

VII. FUTURE SCOPE

The scope of this research can be extended in several directions:

- Implement machine learning models (clustering or classification) to predict user preferences.
- Build a real-time dashboard for dynamic streaming analytics.
- Integrate sentiment analysis from social media reviews for deeper behavioural understanding.
- Explore region-wise or time-based viewing patterns for personalized recommendations.
- Expand to multi-platform behaviour analysis comparing Netflix with competitors like Amazon Prime and Disney+.

VIII. REFERENCES

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